

A Packing of 15 Four-Spaces in $\mathbb{R}^{10}$

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Abstract

We give 15 4-dimensional subspaces of $\mathbb{R}^{10}$ whose smallest squared chordal distance is at least 2.571139. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.559634, obtained by deleting 1 of the 16 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 15 subspaces of dimension 4 in $\mathbb{R}^{10}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.571139 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 10$, $n = 4$. The cell $N = 15$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.559634, obtained by deleting 1 of the 16 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(10 - 4)/10 \cdot 15/14 = 2.571429$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m10n4N15.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 63 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
803481201610074833820367594733346091269895948428598950215937461574116553
276956796365076861768063839197745159748315721682925394128981627977516818
97932741775958694232520363100326267130118858293
```

denominator:

```
312500000000288088770326579541819564260147566975212313670676948441998651
313763360682165931040608061347424745214278049016546432073876648433511328
55948411956996556839638109559615461634609710746
```

B Coordinates

15 blocks of 10 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m10n4N15.txt`.

```
55561930621 -21375573723 487145045295 35509255669 229126728924 -595625171896 96452756774 83551711607 -689941511162 115581023080 130330994650 106850245001
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386349038882 87156122457 -118965356779 -3872382928400414327893 -246186159803 102474568179 -3460005099236570287756 -494191079031 -438060490416 -211170942624
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